

Bootstrap $\rightarrow$ sampling with Replacement.

sample	Feature 1 (Age)	Feature 2 (Income)	Feature 3 (credit score)	Target (Loan Approved)
S ₁	25	40	650	Yes
S ₂	30	60	700	Yes
S ₃	45	35	600	No
S ₄	50	30	500	No
S ₅	28	55	720	Yes

$\rightarrow$ Create 3- bootstrap samples (Size = 5) from the data set.

$\rightarrow$ Train 3 decision tree models using each bootstrap sample.

$\rightarrow$ For a new sample (Age = 32, Income = 50, credit score = 600) determine Prediction from all trees.

$\rightarrow$ complete the final Prediction using majority voting.

① Bootstrap (B ₁)	Bootstrap ₂ (B ₂)	Bootstrap ₃ (B ₃)
S ₁ , S ₂ , S ₂ , S ₄ , S ₅	S ₂ , S ₃ , S ₄ , S ₄ , S ₅	S ₁ , S ₂ , S ₃ , S ₃ , S ₄

② Decision Tree (B ₁)	Bootstrap ₂ (B ₂)	Bootstrap ₃ (B ₃)
S ₁ , S ₂ , S ₂ , S ₄ , S ₅	S ₂ , S ₃ , S ₄ , S ₄ , S ₅	S ₁ , S ₂ , S ₃ , S ₃ , S ₄
Y Y Y N Y	Y N N N Y	Y Y N N N

Age.
 $\begin{array}{l} < 30 \\ > 30 \end{array}$
 Yes No

credit score
 $\begin{array}{l} \geq 650 \\ < 650 \end{array}$
 Yes No

Income
 $\begin{array}{l} \geq 40 \\ < 40 \end{array}$
 Yes No.

$\Rightarrow$ single split DT

3 & 4 Prediction & Voting

Age = 32 ↓ loan Approval: N	Income = 50 ↓ loan Approval: Y	Credit score = 690 ↓ loan Approval: Y.
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Majority voting

N Y Y (Y)

Age = 35 ↓ loan Approval: N.	Income = 45 ↓ Y	Credit score = 750 ↓ Y.
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Majority voting

N Y Y (Y)